

A Reproducible, DNA-Grounded Map of 27 Non-Opioid Analgesic Targets

Promoter switch-threshold (γ) reads sorted into three intervention levers,
with a firewall holding every clinical magnitude open

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jamming-physics.org/analgesic | github.com/reg093-sketch/jamming-physics

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Status. Proposal / preprint. This document is a reproducible scientific *hypothesis*, not medical advice and not a treatment. It identifies and prioritises molecular *targets*; it designs no drug, gives no dose, and asserts no efficacy, potency, or safety result. Those can come only from proper laboratory and clinical validation, which this work does not perform.

Abstract

A single jamming-lattice read — Layer-1 γ , a nearest-neighbour stacking-energy read of the promoter — is taken for 27 nociception (pain-sensing) genes and mapped onto the R19 firing-threshold scale, $|h_{sp}| = (2/3\sqrt{3})\gamma^{1.5}$. The reads are bit-reproducible: $\text{corr}(\gamma, \text{GC}) = 0.998978$ over the 27-target set, drift 0, offline. The central discipline is a firewall: γ reads promoter switch-threshold *structure* only and is never equated with any ion-channel activation voltage, candidate potency, dose, in-vivo selectivity, or clinical effect — all of which are graded **[O]** (open) and asserted nowhere. The map sorts the targets into three intervention levers (reduce the inward current; increase the outward K^+ current; remove the upstream sensitising drive), ranks them by a declared burden weighting, and adds a precision local-anaesthesia map. This is a free, public-benefit, **proposal-only** hypothesis: it designs no molecule and gives no dose, and every clinical magnitude requires external laboratory and clinical work. The complete reproducibility package (engines, cached inputs, gates, frozen hashes, manifest) regenerates every number and verdict deterministically and is released under CC BY 4.0.

1 Scope and the firewall

This paper reads, for 27 genes central to nociception, a single structural number from DNA and uses it to organise a search for non-opioid analgesia. The non-negotiable boundary is a **firewall**: the read γ is a property of a gene's promoter switch-threshold *structure*. It is *not* an ion channel's activation voltage, a drug candidate's potency, a dosing quantity, an in-vivo selectivity, or any clinical effect. Every such magnitude is graded **[O]** (open) and is asserted nowhere in this work. The engine reads structure; it never asserts a clinical number. Equating γ with a clinical magnitude would be a category error and is treated as one throughout.

2 Method: one read per gene

Each gene's human promoter is read by a deterministic engine (a frozen module bundled in the reproducibility package). The read returns γ , a nearest-neighbour stacking-stiffness read of the promoter sequence (nearest-neighbour thermodynamics after SantaLucia, 1998 [1]). γ is then mapped onto the R19

double-well firing-threshold scale,

$$|h_{sp}| = \frac{2}{3\sqrt{3}} \gamma^{1.5}, \quad (1)$$

which locates the gene on a common firing-threshold axis. The read is reproducible ([V]): across the 27-target set the read tracks base composition with $\text{corr}(\gamma, \text{GC}) = 0.998978$ ($n = 27$), and the whole pipeline reproduces bit-for-bit (drift 0, offline). The read fixes *where* a gene sits and *which lever* it belongs to; it never fixes a clinical magnitude.

3 The map: 27 targets across three levers

A pain fibre fires when its inward (depolarising) current overwhelms its outward (hyperpolarising) brake. The map intervenes along three levers, each of which raises the effective firing threshold: **L1** reduce the inward current (Na^+ , Ca^{2+} , proton/ATP-gated); **L2** increase the outward K^+ (K_V7/M -current) brake; **L3** remove the upstream sensitising drive (nerve-growth-factor / TrkA and CGRP signalling). Two further groups are listed for completeness: the developmental master identity switch (PRDM12) and context comparators (opioid and cannabinoid receptors), routed deliberately away from the central reward circuitry that drives opioid addiction and respiratory risk. Table 1 lists all 27 targets with their reads and grades. For the direct-channel levers the lever direction is anchored to cited agents (e.g. the FDA-approved $\text{Na}_V1.8$ blocker) and graded [F]; for the indirect targets the gene is only placed in the map and the receptor/network mechanism is cited biology, never derived, and held [O].

Table 1: The 27-target map. γ is read from each gene's human promoter; $|h_{sp}| = (2/3\sqrt{3}) \gamma^{1.5}$ is its place on the R19 firing-threshold scale. Read grade is [V] and clinical-magnitude grade is [O] for all 27 (stated once, omitted from rows); the per-row grade is the lever/mechanism grade: [F] for the 16 direct-channel targets, [O] for the 11 indirect (L3 sensitiser, master, comparator) targets.

Gene	Channel/Protein	Lever	γ	$ h_{sp} $	Direction (cited)	Gr.
L1 reduce inward current						
CACNA1H	$\text{Ca}_V3.2$	L1	1.6357	0.8052	reduce inward Ca^{2+} (T-type, low-threshold)	[F]
CACNA1B	$\text{Ca}_V2.2$	L1	1.596	0.776 064	reduce inward Ca^{2+} (N-type presynaptic transmitter release)	[F]
ASIC1	ASIC1	L1	1.5161	0.718 522	reduce inward (proton-gated)	[F]
ASIC3	ASIC3	L1	1.4818	0.694 277	reduce inward (proton-gated)	[F]
TRPV1	TRPV1	L1	1.456	0.676 223	reduce inward (heat/capsaicin transducer)	[F]
P2RX3	P2X3	L1	1.4421	0.666 563	reduce inward (ATP-gated, ligand-gated)	[F]
SCN9A	$\text{Na}_V1.7$	L1	1.4014	0.638 545	reduce inward Na^+ (raise AP threshold at the firing gate)	[F]
SCN11A	$\text{Na}_V1.9$	L1	1.3952	0.634 312	reduce inward Na^+ (TTX-R subthreshold/resting)	[F]
SCN10A	$\text{Na}_V1.8$	L1	1.3136	0.579 486	reduce inward Na^+ (TTX-R upstroke)	[F]
TRPM8	TRPM8	L1	1.2719	0.552 112	reduce/modulate inward (cold/menthol transducer)	[F]
SCN3A	$\text{Na}_V1.3$	L1	1.2645	0.547 301	reduce inward Na^+ ($\text{Na}_V1.3$, re-expressed/up-regulated after nerve injury)	[F]
TRPA1	TRPA1	L1	1.2297	0.524 864	reduce inward (irritant/cold transducer)	[F]

continued on next page

Gene	Channel/Protein	Lever	γ	$ h_{sp} $	Direction (cited)	Gr.
L1 (adjacent)						
CACNA2D1	alpha2delta-1 (Ca _v auxiliary subunit)	L1-adjacent	1.4414	0.666 078	reduce inward Ca ²⁺ via alpha2delta-1 (gabapentinoid target)	[F]
L2 increase outward K current						
KCNQ2	K _v 7.2	L2	1.5768	0.762 102	INCREASE outward K ⁺ (open K _v 7.2)	[F]
KCNQ5	K _v 7.5	L2	1.5307	0.728 926	INCREASE outward K ⁺ (open K _v 7.5)	[F]
KCNQ3	K _v 7.3	L2	1.498	0.705 693	INCREASE outward K ⁺ (open K _v 7.3)	[F]
L3 remove sensitising drive						
CALCB	beta-CGRP (ligand)	L3	1.5169	0.719 09	block the CGRP sensitising drive (beta-CGRP ligand)	[O]
NTRK1	TrkA (NGF receptor)	L3	1.5036	0.709 654	block the NGF->TrkA sensitising drive	[O]
CALCA	alpha-CGRP (ligand)	L3	1.4876	0.698 357	block the CGRP sensitising drive (alpha-CGRP ligand)	[O]
NGF	NGF (nerve growth factor, ligand)	L3	1.4688	0.685 16	block the NGF->TrkA sensitising drive (anti-NGF)	[O]
RAMP1	RAMP1 (receptor-activity-modifying protein 1)	L3	1.4249	0.654 673	block CGRP receptor signalling (RAMP specificity component)	[O]
CALCRL	CALCRL (calcitonin-receptor-like receptor)	L3	1.2122	0.5137	block CGRP receptor signalling (CLR receptor component)	[O]
master identity switch						
PRDM12	PRDM12 (transcription factor)	master	1.5955	0.775 699	developmental: specifies the nociceptor lineage (not an acute lever)	[O]
context comparator						
OPRD1	delta-opioid receptor	context	1.476	0.690 204	comparator: delta-opioid	[O]
OPRK1	kappa-opioid receptor	context	1.4351	0.661 716	comparator: peripheral kappa — a non-reward analgesia direction (dysphoria caveat)	[O]
CNR2	CB2 cannabinoid receptor	context	1.3698	0.617 069	comparator: peripheral CB2 — a non-psychoactive direction	[O]
OPRM1	mu-opioid receptor	context	1.2548	0.541 016	comparator: mu-opioid — the REWARD-engaging axis the logic routes AWAY from	[O]

4 Burden-weighted prioritisation

Targets are ranked by burden, unmet need, and druggability under a *declared* editorial weighting — not by their map place. The $\gamma/|h_{sp}|$ place is carried alongside each row but is never folded into the score (firewall: promoter stiffness $\neq$ clinical magnitude). Table 2 ranks the 22 actionable targets; the 5 comparators (OPRM1, PRDM12, OPRK1, CNR2, OPRD1) are held out as non-actionable. The ranking is forced from cited tiers ([F]); it is not an engine output, and it recommends no dose or molecule.

Table 2: Burden-weighted prioritisation of the 22 actionable targets. Score = $0.4 \cdot B + 0.35 \cdot U + 0.25 \cdot D$ over cited 1–5 tiers (B burden, U unmet need, D druggability). The γ/h_{sp} map place is shown alongside but is *never* folded into the score.

#	Gene	Channel/Protein	Lever	B	U	D	Score	Map place
1	SCN10A	Nav _v 1.8	L1	4	5	5	4.6	0.579486
2	NGF	NGF (nerve growth factor, ligand)	L3	5	4	4	4.4	0.68516
3	SCN9A	Nav _v 1.7	L1	4	5	4	4.35	0.638545
4	CALCA	alpha-CGRP (ligand)	L3	5	3	5	4.3	0.698357
5	CALCRL	CALCRL (calcitonin-receptor-like receptor)	L3	5	3	5	4.3	0.5137
6	CACNA2D1	alpha2delta-1 (Ca _v auxiliary subunit)	L1-adjacent	4	4	5	4.25	0.666078
7	NTRK1	TrkA (NGF receptor)	L3	5	4	3	4.15	0.709654
8	CACNA1H	Ca _v 3.2	L1	4	5	3	4.1	0.8052
9	SCN3A	Nav _v 1.3	L1	4	5	3	4.1	0.547301
10	RAMP1	RAMP1 (receptor-activity-modifying protein 1)	L3	5	3	4	4.05	0.654673
11	KCNQ2	K _v 7.2	L2	4	4	3	3.75	0.762102
12	KCNQ3	K _v 7.3	L2	4	4	3	3.75	0.705693
13	CACNA1B	Ca _v 2.2	L1	3	4	4	3.6	0.776064
14	P2RX3	P2X3	L1	3	4	4	3.6	0.666563
15	CALCB	beta-CGRP (ligand)	L3	4	3	3	3.4	0.71909
16	TRPV1	TRPV1	L1	3	3	4	3.25	0.676223
17	KCNQ5	K _v 7.5	L2	3	4	2	3.1	0.728926
18	SCN11A	Nav _v 1.9	L1	3	4	2	3.1	0.634312
19	TRPA1	TRPA1	L1	3	3	3	3.0	0.524864
20	ASIC3	ASIC3	L1	3	3	2	2.75	0.694277
21	TRPM8	TRPM8	L1	2	3	3	2.6	0.552112
22	ASIC1	ASIC1	L1	2	3	2	2.35	0.718522

5 Precision local anaesthesia

A complementary idea makes the block *pain-selective*: pair a nociceptor-selective entry port (a heat/irritant channel such as TRPV1 or TRPA1, open mainly in pain fibres) with a permanently charged firing-threshold raiser that cannot cross resting membranes and so enters only through that open port. The result is a differential block — pain fibres silenced, motor and touch fibres largely spared. The mechanism shape (selective entry → selective block) is [F], anchored to Binshtok, Bean & Woolf (2007) [4]; named agents (e.g. QX-314, chlorprocaine) appear only as experimental anchors. The differential-block ratio, duration, and every other clinical magnitude are [O]. Table 3 lists the entry-port × blocker pairings.

Table 3: Precision local anaesthesia: a nociceptor-selective entry port $\times$ a permanently charged firing-threshold raiser that enters only through that open port. Mechanism shape [F], anchored to Binshtok, Bean & Woolf (2007); every differential-block magnitude is [O].

Entry port	Charged blocker	Rationale (cited)
TRPV1	SCN10A (Na _V 1.8)	Open the nociceptor-restricted TRPV1 port so a permanently-charged Na _V blocker enters only nociceptors, then raise the firing threshold by shutting Na _V 1.8 there — confining the block to the pain fibre.
TRPV1	SCN9A (Na _V 1.7)	Open the nociceptor-restricted TRPV1 port so a permanently-charged Na _V blocker enters only nociceptors, then raise the firing threshold by shutting Na _V 1.7 there — confining the block to the pain fibre.
TRPA1	SCN10A (Na _V 1.8)	Open the nociceptor-restricted TRPA1 port so a permanently-charged Na _V blocker enters only nociceptors, then raise the firing threshold by shutting Na _V 1.8 there — confining the block to the pain fibre.
TRPA1	SCN9A (Na _V 1.7)	Open the nociceptor-restricted TRPA1 port so a permanently-charged Na _V blocker enters only nociceptors, then raise the firing threshold by shutting Na _V 1.7 there — confining the block to the pain fibre.
P2RX3	SCN10A (Na _V 1.8)	Open the nociceptor-restricted P2RX3 port so a permanently-charged Na _V blocker enters only nociceptors, then raise the firing threshold by shutting Na _V 1.8 there — confining the block to the pain fibre.
ASIC3	SCN10A (Na _V 1.8)	Open the nociceptor-restricted ASIC3 port so a permanently-charged Na _V blocker enters only nociceptors, then raise the firing threshold by shutting Na _V 1.8 there — confining the block to the pain fibre.
TRPM8	SCN10A (Na _V 1.8)	Open the nociceptor-restricted TRPM8 port so a permanently-charged Na _V blocker enters only nociceptors, then raise the firing threshold by shutting Na _V 1.8 there — confining the block to the pain fibre.

6 Proposals

Six directional proposals follow from the map. Each is a hypothesis to be tested, not a result.

P1 — Gate.

Intervene at the peripheral nociceptor gate rather than centrally.

P2 — Direction.

Push that gate so as to *raise* the firing threshold (reduce inward current or increase outward current).

P3 — Axis.

Rank effectors by nociceptor selectivity and map place, pursuing the Na_V triad (Na_V1.7/1.8/1.9) as the most selective inward-current axis.

P4 — Shape.

Favour a state-dependent mechanism that lifts the threshold of over-firing fibres while sparing ordinary low-frequency conduction.

P5 — Opioid alternative.

Pursue a peripheral, nociceptor-restricted intervention that is *structurally* non-reward (it does not engage central reward circuitry).

P6 — Differential block.

Pursue pain-selective local anaesthesia via the entry-port $\times$ charged-blocker construction above.

7 Falsification

A map that cannot be wrong is not science. Each proposal names the observation that would refute it: **P1/P2**, no threshold-driven reduction in nociceptive signalling in a controlled assay; **P3**, no nociceptor-selectivity advantage for the Nav triad relative to the other levers; **P4**, a closed/inactivated-state stabiliser cannot lift the threshold while sparing low-frequency conduction; **P5**, a peripheral, nociceptor-restricted intervention nonetheless engages central reward circuitry; **P6**, opening the selective entry port does not confine the charged blocker to nociceptive fibres (no differential block). At the framework level, the open question (**[O]**) — stated, not assumed — is whether the promoter γ read tracks expression-switch behaviour at these loci; if it does not, the Layer-1 grounding of the whole map is falsified.

8 Grading and honesty

Every claim carries one of three grades. **[V] verified / reproducible**: bit-reproducible from public promoter sequence (the 27 γ reads; $\text{corr}(\gamma, \text{GC}) = 0.998978$; drift 0, offline). **[F] forced by the reads**: structural and not chosen (the ordering by $|h_{sp}|$; for the 16 direct-channel targets, the lever placement and direction, anchored to cited agents). **[O] open**: needs external laboratory/clinical data; asserted nowhere. **Every clinical magnitude is [O]**. The discipline is mechanical, not a promise: an *L3-honesty gate* verifies that every upstream-sensitiser target carries an **[O]** cited-biology-never-derived mechanism grade; a *forbidden-claim scanner* fails the build closed if any dosing, synthesis, efficacy, or safety claim appears in the asserted text; and an *irreproducibility ledger* enumerates the open classes with their reasons. Section 9 shows how to run all of these.

9 Reproducibility

This work is built so that every number and verdict can be regenerated from the inputs, offline and bit-for-bit. The reproducibility package accompanying this record (`analgesic_threshold_logic_v2_0_reproducibility.zip`) contains the locked engines, the cited and cached input data, all gate scripts, the frozen output hashes, and a file manifest.

Principle: LOCK $\rightarrow$ Derive $\rightarrow$ Gate. Every input is either a measured/cited quantity (locked) or is derived from locked inputs by a deterministic script — never chosen to fit a target (the no-tuning invariant). Outputs are then verified by counting checks, and a set of output hashes is frozen so that any drift is detected.

Environment. Python 3 with NumPy. All other engine modules are bundled inside the package; no network access is required (promoter sequences are cached in the package and the fetch path is cache-only).

One command.

```
unzip analgesic_threshold_logic_v2_0_reproducibility.zip
cd analgesic_threshold_logic_v2_0
python3 repro/run_all.py
```

Expected final lines:

```
[PASS] determinism -- 10 frozen hashes match (drift 0)
OVERALL: PASS (11/11 checks)
```

What the checks verify. The harness `repro/run_all.py` runs eleven checks and then the determinism freeze:

ID	Stage	Verifies
M1	inherit-reverify	re-derives the inherited nociceptor γ anchor; drift 0
M8	read-22-targets	reads the expanded target set from cached promoter sequence
M8	gate (corr/drift/prov)	$\text{corr}(\gamma, \text{GC}) = 0.998978$ ($n = 27$); drift 0; provenance pinned
M9	three-lever map	builds the 27-target threshold map and lever assignment
M4	channelopathy anchor	cited human channelopathy biology fixes the lever direction
M10	burden-prioritisation	reproduces the declared-weight ranking (Table 2)
M11	L3-honesty (gate)	fail-closed: every L3 sensitiser mechanism is graded [O]
M12	precision-local-anaes	reproduces the entry-port $\times$ blocker map (Table 3)
M5	claim-scan (gate)	fail-closed forbidden-claim scanner (dosing/synthesis/efficacy/safety)
M6	falsification	a named falsifier is present for each proposal
—	determinism	10 frozen output hashes match <code>repro/expected_sha256.json</code> ; drift 0

File manifest. Every shipped file is hashed in `manifest/SHA256SUMS.txt` (94 files). Verify with:

```
cd analgesic_threshold_logic_v2_0
sha256sum -c manifest/SHA256SUMS.txt
```

(all lines report OK).

The open ledger. `IRREPRODUCIBILITY_LEDGER.md` lists every [O] (open) class with its reason — the magnitudes this work deliberately does *not* assert (potency, in-vivo selectivity, efficacy, safety, dose, formulation, and the framework-level expression-switch question). These are the boundary of the claim, recorded honestly rather than hidden.

Canonical edition. A canonical, citable multi-page HTML edition (one self-contained page per target and per section, answer-first, with machine-readable metadata) is included under `docs/analgesic/` and is published at jamming-physics.org/analgesic.

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Released under [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/). Reuse, adapt, and build on the reads and the lever map with attribution; commercial use is permitted, to lower the cost and broaden the search for non-opioid analgesics. All clinical magnitudes remain the reusing party’s own scientific, ethical, and regulatory responsibility.

References

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A Reproducibility package layout

```

analgesic_threshold_logic_v2_0/
  README.md  CONSTITUTION.md  VP_SPEC_v1_8.md  IRREPRODUCIBILITY_LEDGER.md
  CHANGELOG_v2_0_multipage.md  build_multipage_analgesic.py
  paper/    analgesic_threshold_logic_v2_0.tex  .pdf  build_paper_tex.py
  docs/
    analgesic/index.html                hub (answer-first + contents + cross-links)
    analgesic/<slug>/index.html          40 self-contained pages: how-to-read,
                                         5 levers, 27 per-target, 7 sections
    assets/css/site.css  robots.txt  sitemap.xml  llms.txt  analgesic/_meta.json
  repro/
    _engine/                          locked engines (dna_interpreter, vp_neuro_engine)
    _inherited_data/                  inherited gamma + cached promoter sequences
    01-inherit-reverify/  02-read-nav-channels/  03-threshold-map/
    04-channelopathy-anchor/  05-intervention-logic/ (forbidden_claim_scan.py)
    06-falsification/  10-burden-prioritisation/  11-13-honesty/
    12-precision-local-anaesthesia/
    run_all.py                      expected_sha256.json  (10 frozen hashes)
  manifest/SHA256SUMS.txt            every shipped file hashed
  reports/                           gate snapshots

```